

Daily Research Digest - 2026-04-08

Topics: galaxy clusters, galaxies, gravitational lensing, dark matter

Synthesis

Executive overview

Recent submissions span galaxy studies, dark matter investigations, and gravitational lensing insights. A notable focus is on improving redshift estimates for faint galaxies to enhance cosmological data interpretation, while dark matter research delves into mixed fuzzy-cold models and explores novel detection methods like the Einstein Telescope and future MeV telescopes.

Galaxy clusters

There are no recent preprints focused specifically on galaxy clusters in this batch of papers. Future updates may cover this area more comprehensively as new studies emerge.

Galaxies

The papers focus on improving photometric redshifts through deep spectroscopic surveys using DESI and examining the stellar populations in dusty early-type galaxies (ETGs). The "Deep Spectroscopy with DESI" paper demonstrates that DESI can efficiently measure redshifts for extremely faint galaxies, crucial for enhancing the accuracy of large-scale cosmological surveys. Meanwhile, "Stellar Population Characterisations in nearby, dusty Early-Type Galaxies" reveals that many dusty ETGs exhibit signs of young or intermediate-age stellar populations, suggesting ongoing or recent star formation.

Gravitational lensing

There are no direct contributions to gravitational lensing studies from this batch of preprints. However, advancements in galaxy redshift measurements and dark matter models (e.g., mixed FDM-CDM) may indirectly impact future lensing analyses by improving the precision of cosmological parameters and understanding substructure effects.

Dark matter

The papers explore diverse aspects of dark matter, including axion signals from supernovae, fuzzy and cold dark matter mixtures, and parity testing in galactic surveys. "Hunting Dark Matter with the Einstein Telescope" discusses using future gravitational wave detectors to probe scenarios where primordial black holes might form the bulk of dark matter. The paper on mixed FDM-CDM highlights significant differences between N-body simulations and full Schrödinger-Poisson dynamics, indicating that velocity field effects are critical for understanding Lyman- α observables. Additionally, "Testing parity with composite-field spectra" employs advanced statistical methods to search for parity violation in galaxy surveys without finding conclusive evidence.

Cross-cutting themes

A recurring theme across these papers is the application of advanced observational techniques and computational models to refine our understanding of cosmological parameters and dark matter properties. Improvements in spectroscopic redshifts (galaxies) can enhance gravitational lensing studies, while detailed simulations of dark matter dynamics inform galaxy formation and evolution theories.

Papers to read first

- Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration - This paper showcases the efficiency of DESI in measuring faint galactic redshifts, critical for large-scale cosmological surveys. - Lyman- α Forest Signatures of Mixed Fuzzy and Cold Dark Matter -

Demonstrates that wave-mechanical effects in mixed dark matter models leave distinct signatures in Lyman- α observables.

- Hunting Dark Matter with the Einstein Telescope - Proposes using future gravitational wave detectors to explore scenarios where primordial black holes could form dark matter.

Open questions and follow-ups

The abstracts suggest several unresolved areas: 1. Galaxies: The origin of dust in ETGs remains partially unclear, needing further exploration. 2. Dark Matter: Mixed FDM-CDM models require more detailed observational validation to confirm the differences highlighted by simulations. 3. Gravitational Lensing: Future lensing studies could benefit from improved redshift measurements and refined dark matter substructure understanding.

These areas provide fertile ground for future research, particularly in refining cosmological models and improving observational techniques.

Paper Summaries and Figures

1. Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration

Focus: galaxies

Authors: Biprateep Dey, Jeffrey A. Newman, Tianqing Zhang, J. Aguilar, S. Ahlen, A. Anand, B. Andrews, S. Bailey, D. Bianchi, D. Brooks, F. J. Castander, T. Claybaugh, A. Cuceu, K. S. Dawson, A. de la Macorra, J. Della Costa, Arjun Dey, P. Doel, S. Ferraro, A. Font-Ribera, E. Gaztañaga, Satya Gontcho A Gontcho, D. Gruen, G. Gutierrez, J. Guy, H. K. Herrera-Alcantar, K. Honscheid, M. Ishak, R. Joyce, R. Kehoe, D. Kirkby, T. Kisner, A. Kremin, O. Lahav, M. Landriau, L. Le Guillou, A. Leauthaud, M. E. Levi, M. Manera, P. Martini, J. McCullough, A. Meisner, R. Miquel, J. Moustakas, A. D. Myers, J. Myles, S. Nadathur, N. Palanque-Delabrouille, W. J. Percival, F. Prada, I. Pérez-Ràfols, G. Rossi, L. Samushia, E. Sanchez, D. Schlegel, M. Schubnell, H. Seo, J. Silber, D. Sprayberry, G. Tarlé, B. A. Weaver, N. Weaverdyck, R. H. Wechsler, R. Zhou, H. Zou

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Summary: Deep spectroscopic samples can be used to improve photometric redshift (photo-z) estimates and reduce uncertainties on redshift distributions. Such improvements can increase the cosmological constraining power of large imaging-based experiments such as the Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST) and mitigate what may be a limiting systematic effect. We present results from the DESI-Deep pilot" program, which was designed to assess the capability of the Dark Energy Spectroscopic Instrument (DESI) on the 4m Mayall telescope to measure redshifts of galaxies as faint as expected lensing samples for early LSST data ($m_i \leq 24.5$). We find that DESI is remarkably efficient at this task, with redshift success rates comparable to the results of observations from 10m-class telescopes with only 2 longer integration time (rather than 8 longer as would be expected from aperture-area scaling), while simultaneously achieving 30 times larger multiplexing. We also find that the signal-to-noise ratio of the spectra scales as expected for background-limited observations even for the longest exposure times (7 hours) and faintest targets in the program. These results demonstrate that DESI could provide the definitive redshift sample for the early years of LSST with a modest investment of observing time. Based upon the results of this program, we provide updated predictions for the time required to collect benchmark samples for photo-z training and calibration using a variety of spectroscopic facilities. Finally, we describe a potential "DESI-Deep" survey designed to train and calibrate photo-z's for imaging experiments, and provide forecasts of its impact on cosmological inference.

Page-by-page summaries

Page 1: The paper outlines the potential of DESI for enhancing photometric redshift (photo-z) accuracy by providing precise spectroscopic data for faint galaxies, comparable to that from larger telescopes but with greater efficiency. It highlights the effectiveness of DESI in measuring redshifts for early LSST data and proposes a "DESI-Deep" survey aimed at improving photo-z training and calibration, which could significantly impact cosmological inference.

Page 2: The page emphasizes the critical role of large, unbiased spectroscopic redshift datasets in improving both training and calibration of photometric redshifts. It discusses how these datasets are essential for refining galaxy SED templates and reducing uncertainties in empirical methods, and highlights the necessity endorsed by various community planning exercises. The text also stresses that magnitude-limited spectroscopic samples, rather than color-selected ones, are crucial to avoid biases and accurately represent the target galaxy population.

Page 3: DESI's high multiplexing capabilities and large field-of-view enable deep spectroscopic samples that enhance existing dark energy experiments; a pilot program targeting fainter galaxies (mi

■24.05) assesses DESI's performance for future photometric redshift training and calibration, with observations conducted in specific sky patches to leverage multi-wavelength data.

Page 4: The page details the selection and observation process for DESI galaxies, applying quality cuts and dust extinction corrections to HSC catalog data, and employing a dithering strategy during observations to enhance data reliability. Targets were prioritized based on magnitude ranges, ensuring a uniform distribution across magnitudes while optimizing exposure times for fainter objects.

Page 5: Page 5 of "Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration" highlights DESI's capability to secure reliable redshifts for faint galaxies, demonstrating performance across quiescent and star-forming galaxy spectra. The methods section details a visual inspection (VI) campaign used to validate redshift measurements from the DESI spectroscopic pipelines, assigning quality scores based on spectral features visibility to determine secure redshifts.

Page 6: Page 6 of "Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration" presents two figures: Figure 1 illustrates the spatial distribution and exposure times of 4863 observed objects in DESI-XMMLSS and DESI-COSMOS fields, highlighting empty central regions reserved for other observations; Figure 2 shows the i-magnitudes and i-fiber-magnitudes distributions, indicating that objects with i-magnitude > 23 were given higher priority, leading to longer exposure times but fewer total objects observed.

Page 7: Page 7 discusses the adjusted redshift measurement efficiency for DESI, focusing on separating redshift measurement failures due to intrinsic high-redshift positions ($z > 1.6$) from those with insufficient signal-to-noise ratios within covered spectral ranges. Two methods—empirical using COSMOS2020 data and analytic based on LSST Science Book relations—are employed to estimate the number of galaxies at $z > 1.6$, though both have limitations in high-redshift regimes. Additionally, the page introduces an empirical method for calculating signal-to-noise ratios (SNR) independent of the DESI spectroscopic pipeline to assess how SNR changes with exposure time and source brightness, aiming to identify potential instrumental improvements.

Page 8: Figure 4 illustrates spectra from four example galaxies in the DESI-Deep pilot sample, showcasing diverse galaxy types with reliable redshift determinations; quiescent and star-forming galaxies are differentiated by dominant absorption or emission lines, respectively, while high signal-to-noise ratios highlight the instrument's capability to measure fainter targets effectively.

Page 9: Page 9 discusses the development of a model for predicting the probability of obtaining successful redshift measurements using DESI spectroscopy data, based on exposure time and source brightness. The model employs logistic regression with an adjusted magnitude parameter, yielding probabilities through a sigmoid function. Additionally, it highlights issues with relying solely on $\Delta\chi^2$ thresholds for assessing redshift reliability, favoring visual inspection labels (VI) instead for quality assessment in the DESI-Deep pilot survey context.

Page 10: The page discusses the evaluation of DESI's redshift measurement success rate using VI quality flags and compares it with DEEP2/3 and zCOSMOS surveys. DESI achieves a consistent success rate greater than 75% for brighter objects with exposure times around 1.5 hours, and matches DEEP2/3 success rates with approximately 2 hours of exposure time, indicating effective performance across different magnitudes and exposure durations.

Page 11: The page discusses how DESI spectroscopy redshift measurement success rates increase with longer exposure times, particularly noting that brighter objects require less exposure time compared to fainter ones. It also introduces a model predicting the redshift success rate based on source magnitude and exposure time, showing that DESI follows an expected SNR scaling of $T_{\text{exp}}^{0.5}$.

Page 12: The page presents Figure 7, which illustrates the redshift measurement success rate for DESI-Deep as a function of i-band magnitude across different exposure times, showing that DESI matches or exceeds the success rates of DEEP2 and DEEP3 surveys at approximately 2 hours of exposure time. The text also introduces an equation to model exposure time based on magnitude and success probability, indicating that this model can predict future survey requirements effectively.

Page 13: Page 13 of "Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration" includes Figure 8, which illustrates the redshift measurement success rate for the DESI-Deep pilot sample as a function of i-band fiber magnitude across various exposure times. The figure compares

DESI's performance to DEEP2/3 and zCOSMOS-Bright surveys, showing that DESI achieves comparable success rates at about 2 hours of exposure time despite higher throughput efficiency due to advancements in its instrumentation design.

Page 14: Page 14 discusses the redshift measurement success rate for the DESI-Deep pilot sample as a function of exposure time and i-band magnitude, highlighting an increase in success rate with longer exposure times. It also compares the astrometric accuracy between Keck/DEIMOS and DESI, noting that DESI's higher precision reduces light losses compared to DEEP2/3 observations due to narrower slits. Additionally, the page outlines differences in exposure time strategies between DESI and DEEP2/3 surveys, suggesting these differences partly explain variations in redshift measurement efficiency but are not entirely sufficient to account for all observed discrepancies.

Page 15: Page 15 of "Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration" presents Figure 10, which illustrates the impact of adjusting redshift success rates for objects with $z > 1.6$ based on i-band magnitude and exposure time. The figure compares measured success rates with adjustments using COSMOS2020 data and analytic fits, showing that even for fainter objects, most failures are due to objects at $z \leq 1.6$.

Page 16: Page 16 details how DESI's sky subtraction method, leveraging non-local modeling and PCA-based adjustments, significantly reduces systematic errors, allowing deeper observations while maintaining expected SNR scaling. Additionally, DESI's broader wavelength coverage (from 3600 to 9800 Å) enhances redshift measurement efficiency compared to previous surveys like DEEP2/3, enabling the detection of spectral features at higher redshifts up to $z \approx 1.6$.

Page 17: Page 17 of the paper presents Figure 12, which shows that DESI's empirical SNR measurements align with background-limited expectations across various exposure times and magnitudes. Additionally, DESI's high spectral resolution, especially in the infrared range (6560–9800 Å, R~4000), allows for the clear separation of narrow features like the [O II] doublet at $z > 0.76$, enhancing its redshift determination capabilities compared to lower-resolution instruments such as DEEP3 and VIMOS.

Page 18: Page 18 of "Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration" discusses the high redshift efficiency of DESI compared to previous surveys like Keck/DEIMOS, attributing this partly to its multi-exposure dithering strategy which averages out instrumental systematics. The section recalculates the telescope time required for a benchmark spectroscopic survey designed for photo-z training using DESI's performance metrics, providing updated estimates for various new facilities and instrument designs.

Page 19: The page discusses three scenarios—Field-Limited, FoV-Limited, and Multiplex-Limited—defining the observing time required to meet survey requirements with DESI-like efficiency. It highlights that future spectroscopic facilities could achieve significant reductions in necessary dark years (~3×) compared to previous estimates, thus making photo-z training and calibration for faint galaxies feasible. Additionally, it outlines a potential "DESI-Deep Survey" design targeting magnitude-limited galaxies ($18 \leq m_i \leq 24.5$), aiming to provide transformative data for cosmological constraints and extragalactic science.

Page 20: Table 1 on page 20 compares the total effective exposure times required for photometric redshift training and calibration using various spectroscopic facilities, with DESI-like efficiency assumptions. The table highlights significant differences in observational time requirements across telescopes/instruments like DESI, Keck/DEIMOS, TMT/WFOS, and ELT/MOSAIC Optical/NIR, noting that proposed facilities are included but not yet operational.

Page 21: Page 21 of "Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration" outlines the strategy for observing approximately 20,000 objects across two fields to calibrate redshift distributions in cosmological analyses. The aim is to achieve high precision in characterizing mean redshifts ($\bar{z}$) and their spreads within tomographic bins, using deep photometry from ideal fields like COSMOS and XMM-LSS. Observations are planned with 1000-second exposures in a rosette pattern to suppress instrumental systematics, targeting an overall success rate of at least 75% for objects down to i-band magnitude 24.5.

Page 22: The page discusses forecasts for the Dark Energy Task Force Figure of Merit (FoM) in LSST 3×2pt analyses under two extreme scenarios of redshift failures, using a Fisher forecasting framework.

The success rate of redshift measurements is crucial for determining the effective number density of source galaxies across five tomographic bins, impacting the overall constraining power of cosmological parameters. FoM forecasts account for both uniform and concentrated (at highest redshifts) failure scenarios, with assumptions based on high SNR photometry enabling robust galaxy characterization.

Page 23: The page discusses the forecasting exercise linking spectroscopic success rates to cosmological FoM degradation, showing that an 80% success rate results in FoM values between 0.96 and 0.64 for LSST Year-1 analysis, corresponding to a 2% to 25% increase in cosmological parameter error bars. It also highlights the feasibility and robustness of the DESI-Deep survey in providing critical spectroscopic samples for future photometric redshift training.

Page 24: Figure 15 illustrates the relationship between DESI exposure time, redshift success rate, and cosmological constraining power (FoM/FoMfiducial) for LSST Year-1 and Year-10-like analyses. The figure contrasts scenarios with uniform and worst-case (highest-redshift only) spectroscopic failures, showing that achieving an 80% success rate in redshift measurements results in a FoM degradation between 4% and 36%, which translates to an increase in cosmological parameter error bars of 2% to 25%.

Page 25: Page 25 details the extensive funding and institutional support for DESI-related research, including contributions from multiple U.S. governmental agencies, international foundations, and member institutions. It also acknowledges the use of various facilities and software tools in conducting the research and emphasizes the ethical guidelines followed regarding AI assistance in manuscript preparation.

Page 26: Additional observations in the DESI-Hercules field, 7,370

Page 27: Figure 17 illustrates the measurement success rate for the DESI-Deep pilot sample as a function of i-band magnitude across different effective exposure times, showing lower standard errors but slightly reduced success rates with the inclusion of DESI-Hercules data compared to Figure 7. At approximately 2-hour exposure times, DESI's success rate aligns with that of the DEEP2 and DEEP3 surveys despite these changes.

Page 28: Figure 18 illustrates the comparison between empirically measured and expected Signal-to-Noise Ratio (SNR) as a function of i-fiber-magnitude for various exposure times in the DESI-Deep pilot sample, including data from the DESI-Hercules field. The observed SNR scaling is slightly worse at fainter magnitudes compared to previous observations, which is attributed to the non-optimal status of the instrument during the HERCULES observations.

Page 29: Page 29 of the research paper lists references that support the use of DESI spectroscopy for training and calibrating photometric redshifts, citing a variety of studies from different years and journals that underpin the methodologies and technologies relevant to this field.

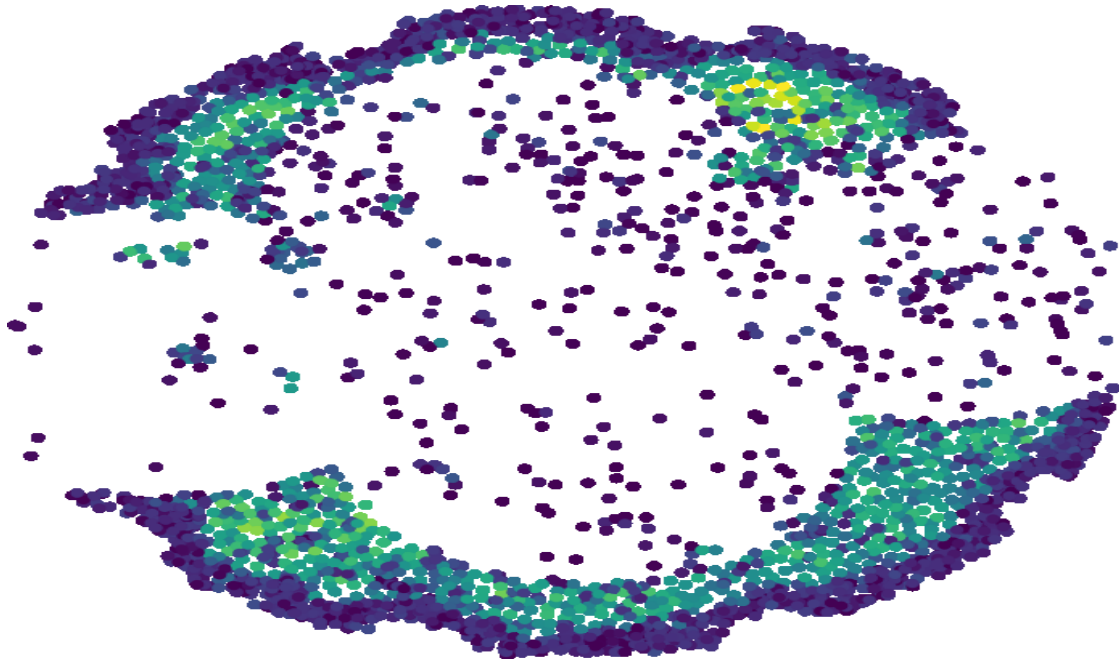
Page 30: Page 30 of the research paper "Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration" consists primarily of a list of references cited throughout the document, covering a range of topics from astronomical surveys to data analysis techniques relevant to photometric redshift studies.

Page 31: Page 31 of the research paper "Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration" is dedicated to listing references that underpin the methodologies, data analysis tools, and related studies utilized in the context of photometric redshift training and calibration using DESI spectroscopic data.

Page 32: Page 32 of the paper "Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration" lists all contributing authors along with their affiliations from various institutions worldwide, highlighting the collaborative nature of the research.

Page 33: Page 33 of the paper lists the affiliations of the authors contributing to the research on DESI spectroscopy for photometric redshift training and calibration, representing institutions from various countries including Canada, Spain, South Korea, Georgia, USA, and China.

Selected figures



2. Hunting Dark Matter with the Einstein Telescope

Focus: dark matter

Authors: A. J. Iovino, M. Maggiore, N. Muttoni, A. Riotto

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Summary: Too light primordial black holes evaporate and are therefore strongly constrained by various bounds, e.g. Cosmic Microwave Background distortion. However, if they are formed strongly clustered, the corresponding haloes may collapse in heavier black holes which may form the entirety of the dark matter of the universe. The indirect signal of such scenario is the production of a flat stochastic background of gravitational waves which is detectable by the Einstein Telescope.

Page-by-page summaries

Page 1: The abstract posits that primordial black holes, if formed in strongly clustered haloes and subsequently collapsing into heavier black holes, could constitute all of the universe's dark matter, producing a detectable flat stochastic background of gravitational waves with the Einstein Telescope.

Page 2: The page introduces Primordial Black Holes (PBHs) as potential Dark Matter candidates in the asteroid mass range and discusses their gravitational-wave signals detectable by experiments like LISA and the Einstein Telescope (ET). It highlights that while ET is sensitive to frequencies corresponding to much lighter PBHs, which have since evaporated, it may still indirectly provide insights into DM if PBHs exhibit strong clustering beyond Poisson expectations at their time of formation.

Page 3: This page introduces the concept of clusterogenesis, where light Primordial Black Holes (PBHs) can cluster and merge into heavier PBHs that could constitute dark matter, while potentially producing a stochastic spectrum of second-order gravitationally induced gravitational waves (SIGWs) detectable by the Einstein Telescope (ET). The section sets up the technical framework using the curvature perturbation $\zeta(x)$ and the compaction function $C(x)$, detailing their definitions and relationships crucial for understanding PBH formation.

Page 4: Page 4 discusses the compaction function under spherical symmetry and its relation to density perturbations, introduces the PBH two-point correlation function using discrete PBH centers and their overdensity, and outlines how peaks in curvature perturbation correspond to peaks in the compaction function, with a Taylor expansion approximation for large amplitude peaks.

Page 5: The page discusses the compaction function $C(r)$ and its linear counterpart $C_{\text{lin}}(r)$, establishing that their maxima coincide under certain conditions, specifically for type I primordial black holes (PBHs). It further links the thresholds of the compaction function to those of fully non-Gaussian curvature perturbations, introducing a functional form $\zeta(x) = F[\zeta_g(x)]$ to describe these perturbations and their probabilities. The correlation of the thresholds is derived using threshold statistics, providing a method for studying PBH correlations through the lens of non-Gaussian curvature perturbations.

Page 6: where ζ_c is the threshold. The key point is the conservation of probability $P[\zeta]d\zeta = P[\zeta_g]d\zeta_g$, $P[\zeta(0), \zeta(r)]d\zeta(0)d\zeta(r) = P[\zeta_g(0), \zeta_g(r)]d\zeta_g(0)d\zeta_g(r)$, (2.17) from which we deduce that the correlation function of the fully non-Gaussian curvature perturbation can be derived from the Gaussian counterpart, provided that one appropriately shifts the threshold $\zeta_{cg} = F^{-1}[\zeta_c]$. (2.18) For the example, f

Page 7: the Gaussian counterpart through the conservation of the probability, the Gaussian threshold ζ_{cg} tends to zero. This simple reasoning confirms the result of Ref. [31], that when the threshold of the non-linear ζ is in the exponential tail region of the probability, the correlation loses its dependence on the threshold itself. What values of ζ_{cg} should we use? The present abundance of DM in the fo

Page 8: Page 8 discusses the spatial exclusion effect in Primordial Black Hole (PBH) formation, establishing that PBHs cannot form arbitrarily close to each other; thus, the correlation function $\xi(r)$ must be evaluated at a minimum distance $r_{\text{exc}} \geq 2r_m$ to avoid overlapping peak profiles. The average number of PBHs in a comoving volume is calculated through equation (3.2), indicating significant deviation from Poisson distribution when clustering effects dominate over discreteness noise.

Page 9: The page discusses the clustering length r_{cl} and its relevance in estimating the average number of Primordial Black Holes (PBHs) within a cluster, given by N , and their potential to form virialized halos during the radiation phase through non-linear overdensities. The spherical collapse model is utilized to describe this process, indicating that PBH clusters can decouple from the Hubble flow in the radiation era, potentially collapsing into more massive BHs if they meet the Hoop conjecture criteria.

Page 10: This page discusses constraints and requirements for primordial black holes (PBHs) as dark matter candidates, including the Hoop conjecture and the necessity of an extended power spectrum with $k_{\text{max}}/k_{\text{min}} \sim 10^{6.5}$ to satisfy these conditions without PBH evaporation preceding cluster formation times. It also specifies that for $M_{\text{PBH}} \sim 10^{-19} M_{\odot}$, the width of the power spectrum (Δ) should be between 6.5 and 8 for PBHs to account for dark matter, which can be achieved in curvaton models and USR models with prolonged constant roll phases.

Page 11: Page 11 discusses the spectral density of induced gravitational waves (GWs) associated with primordial black hole (PBH) production, providing a formula (4.1) for this density under radiation domination and noting that effects like the Electroweak phase transition and primordial non-Gaussian corrections are neglected for simplicity. The page also mentions that future work will address these neglected effects.

Page 12: The page discusses the detectability of second-order gravitational wave (GW) signals using different configurations of the Einstein Telescope (ET), including triangular (ETTR), 2L-aligned (ETLLa), and 2L-misaligned (ETLLm). It compares ET's sensitivity with other detectors like LISA, LLR, eLO, and AION-km, under a Signal-to-Noise ratio of 10 over a year of observation, assuming specific benchmark values for the non-Gaussian parameter A_b .

Page 13: The page discusses how the alignment angle β between two L-shaped Einstein Telescope detectors impacts sensitivity to stochastic backgrounds, with maximal sensitivity at $\beta = 0^\circ$ (ETLLa) and minimal sensitivity at $\beta = 45^\circ$. A small misalignment of about 2.5° (ETLLm) is chosen to balance sensitivity to both stochastic backgrounds and compact binary coalescences. The paper considers three configurations for sensitivity analysis, highlighting the potential for detecting PBH-generated SIGW in the ET range, which could indirectly reveal the presence of dark matter.

Page 14: Page 14 lists references that support the exploration of primordial black holes as dark matter candidates and the potential for detecting their gravitational waves using instruments like LISA and the Einstein Telescope, highlighting studies on clustering effects and observational prospects.

Page 15: The page lists several key studies examining the clustering behavior and formation mechanisms of primordial black holes, emphasizing their potential implications for dark matter and cosmological observations, including gravitational wave signals and microlensing constraints. References highlight the effects of non-Gaussian initial fluctuations on these phenomena, contributing to a broader understanding of early structure formation in the universe.

Page 16: Page 16 of "Hunting Dark Matter with the Einstein Telescope" lists references focusing on theoretical aspects of inflationary models, primordial black holes (PBHs), and their potential connections to dark matter. Key topics include PBH formation in curvaton models, the gravitational wave background from primordial density perturbations, and non-Gaussianity in cosmological scenarios.

Page 17: The references on this page highlight recent studies exploring the relationship between non-Gaussian scalar perturbations and induced gravitational waves, considering various aspects such as cosmological signatures, pulsar timing array constraints, and implications for dark matter isocurvature. These studies collectively aim to deepen our understanding of primordial gravitational wave backgrounds and their potential origins from scalar-induced perturbations.

Page 18: Page 18 lists references discussing various gravitational wave detection methods and their applications in searching for dark matter, including scalar-induced gravitational waves, LISA mission prospects, atom interferometer observatories like AION, and the sensitivity of these instruments to gravitational waves and ultralight dark matter. Additionally, it mentions studies on the Einstein Telescope's design and its potential scientific impact.

3. Lyman- α Forest Signatures of Mixed Fuzzy and Cold Dark Matter

Focus: dark matter

Authors: Yourong Frank Wang

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Summary: We investigate Lyman-alpha forest flux statistics in mixed fuzzy dark matter (FDM) and cold dark matter (CDM) cosmologies using the Fluctuating Gunn-Peterson Approximation (FGPA) applied to hybrid Schrödinger-Poisson and N-body simulations. We evolve the dark matter distribution from $z = 120$ to $z = 2$ for an axion mass ($m_{22} = 0.01$) and FDM fraction ($f_A = 0.1$), and compare two realizations with identical initial conditions: one evolved with a particle-only approximation and one with full wave-mechanical dynamics. We find that, despite near-degeneracy in the nonlinear matter power spectrum, the corresponding Ly α flux power spectra differ at the 10 percent level on intermediate scales. This discrepancy arises from a strong suppression of small-scale velocity power in the Schrödinger-Poisson evolution, which is not captured by N-body treatments with matched initial transfer functions. As a result, the flux statistics cannot be fully characterized by the matter power spectrum alone, but depend sensitively on the dynamical evolution of the velocity field. These results demonstrate that wave-mechanical effects in FDM leave distinct kinematic imprints in Ly α observables beyond those associated with initial-condition suppression. While our analysis is based on an idealized FGPA framework, it isolates a mechanism by which mixed dark matter models can break degeneracies present in standard structure-based probes, motivating further investigation with full hydrodynamical simulations.

Page-by-page summaries

Page 1: The page introduces a study that investigates the Lyman- α forest flux statistics in mixed fuzzy and cold dark matter cosmologies using hybrid Schrödinger-Poisson and N-body simulations. It highlights that despite similarities in nonlinear matter power spectra, significant ($\sim 10\%$) differences in

Ly α flux power spectra arise due to distinct small-scale velocity dynamics between wave-mechanical and particle-only approximations, indicating sensitivity of flux statistics to dynamical evolution beyond initial conditions.

Page 2: The page discusses how Fuzzy Dark Matter (FDM) dynamics can be modeled using a coherent wavefunction governed by the Schrödinger–Poisson equation, with distinct impacts on cosmic structures at different redshifts. The Lyman- forest, a series of absorption features in quasar spectra due to neutral hydrogen, is highlighted as a sensitive probe for detecting small-scale power suppression characteristic of FDM models. The paper also outlines the use of the AxioNyx code and the Fluctuating Gunn–Peterson Approximation (FGPA) to model these effects while isolating gravitational impacts from complex thermodynamics in the intergalactic medium.

Page 3: The page details the simulation methods for modeling mixed fuzzy and cold dark matter, including the initialization of axion wavefunctions and CDM particles, and the use of AxioNyx to evolve these components with high-order accuracy. The text also describes the time-stepping strategy necessary to maintain stability across different timescales and introduces the process of generating mock Ly- forest transmission skewers using a Fluctuating Gunn–Peterson Approximation forward model for comparison among Λ CDM, mixed dark matter N-body, and full Schrödinger–Poisson dynamics simulations.

Page 4: Page 4 discusses the simulation techniques used for modeling mixed fuzzy dark matter (FDM) and cold dark matter (CDM), including baryonic pressure smoothing with a Gaussian filter to damp small-scale fluctuations. It also details the method for extracting velocity information from the complex phase of the wavefunction in a grid-based Schrödinger–Poisson solver, and introduces the construction of the total line-of-sight velocity by density-weighting CDM and FDM components based on their local fractions.

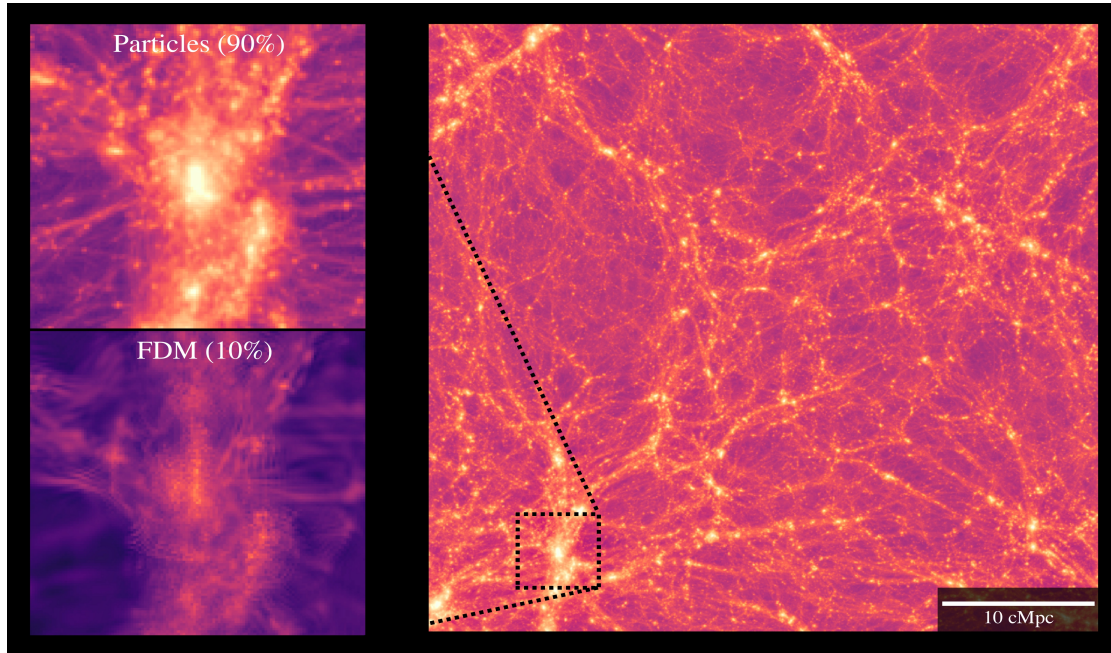
Page 5: The page discusses the computational methods used to model Lyman- forest signatures in a mixed fuzzy and cold dark matter scenario, detailing the evaluation of recombination coefficients, thermal broadening effects, redshift-space mappings, optical depth calculations, and flux transmission. It highlights that these computations remain qualitatively consistent under moderate variations of initial thermal parameters and emphasizes the sensitivity of Lyman- forest observations to differences in neutral hydrogen distribution and coherent line-of-sight velocity fields influenced by mixed dark matter components.

Page 6: The page discusses simulations comparing mixed dark matter (MDM) models with cold dark matter (CDM) in generating Lyman- forests. At $z=2$, the MDM models show a contained large-scale flux power excess relative to Λ CDM, but stronger suppression on smaller scales, particularly around $k \sim 0.1 \text{ s/km}$, highlighting robust FDM signatures against IGM thermal uncertainties. The full Schrödinger–Poisson treatment suppresses velocity power more significantly than the N-body approximation due to the density-weighted composite velocity field, where the axion component dominates in moderate-overdensity regions, leading to substantial velocity suppression beyond what is expected from cosmic mean fractions alone.

Page 7: The page discusses how mixed Fuzzy Dark Matter (FDM) and Cold Dark Matter (CDM) cosmologies suppress the Lyman- α flux power spectrum relative to Λ CDM at scales of $k \sim 0.05\text{--}0.2 \text{ s/km}^{-1}$, with a full Schrödinger–Poisson treatment showing stronger suppression than an N-body approximation due to differences in the velocity field's kinematic structure. The results indicate that N-body-based emulators may underestimate Lyman- α suppression, which has implications for future observational constraints from surveys like DESI and Euclid.

Page 8: Page 8 of "Lyman- α Forest Signatures of Mixed Fuzzy and Cold Dark Matter" emphasizes the necessity for accurate modeling of Lyman- observables to capture the kinematic effects of wave-mechanical dynamics in mixed dark matter cosmologies. The author acknowledges significant contributions from collaborators, computational resources, and simulation tools used throughout the research.

Selected figures



4. Testing parity with composite-field spectra of BOSS and DESI luminous red galaxies

Focus: galaxies

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Summary: Detection of parity violation on cosmological scales would have profound implications for fundamental physics. Motivated in part by recent measurements of parity-odd four-point correlation functions in BOSS and DESI luminous red galaxy samples, which probe parity violation in the scalar sector, we present the first measurement of parity-odd kurto spectra in spectroscopic galaxy survey data. We analyse two composite-field spectra, P_{22} (vector--pseudo-vector) and P_{31} (scalar--pseudo-scalar). Compared with parity-odd four-point correlation function analyses, the kurto-spectrum formalism performs physically motivated compression on the trispectrum into a substantially lower-dimensional data vector, allowing direct estimation of covariance matrices from mock catalogues and reducing sensitivity to covariance-modelling systematics. Using null-hypothesis χ^2 tests and cross-patch consistency checks, we find no evidence for a cosmological parity-violating signal in either survey. We examine the impact of the adopted mock catalogues and find that the high-fidelity mocks provide a better match to the data of both surveys than the approximate mocks. The DESI DR1 measurements exhibit a scatter smaller than that of BOSS DR12 by about a factor of four, consistent with the improved statistical precision expected from the higher tracer number density. Future DESI data releases, with larger volume and number density, together with larger suites of high-fidelity mocks, can enable significantly sharper tests of parity violation using kurto spectra.

Page-by-page summaries

Page 1: The authors present the first measurement of parity-odd kurto spectra in BOSS and DESI luminous red galaxy samples, using composite-field spectra $P_{2 \times 2}$ (vector–pseudo-vector) and $P_{3 \times 1}$ (scalar–pseudo-scalar), which compress trispectrum data into lower-dimensional vectors. No evidence for cosmological parity-violating signals is found in either survey through null-hypothesis χ^2 tests and

cross-patch consistency checks, with DESI DR1 showing a smaller scatter compared to BOSS DR12 due to higher tracer number density.

Page 2: The page introduces the significance of detecting parity-odd cosmological signals in identifying new fundamental physics or astrophysical sources, focusing on scalar sector violations observable through large-scale structure surveys like BOSS and DESI. It highlights that parity violation first manifests at leading order in the four-point correlation function (4PCF) and trispectrum, emphasizing the challenges of measuring these high-dimensional statistics.

Page 3: Page 3 summarizes challenges in detecting parity violation through composite-field (CF) spectra analysis of luminous red galaxies from BOSS and DESI. It highlights issues with covariance estimation methods and data-mock mismatches that can bias detection results, while emphasizing the potential of CF spectra to robustly compress N-point correlation functions into two-point statistics, facilitating more reliable covariance estimation and parameter constraints without relying heavily on Gaussian-field assumptions.

Page 4: The page outlines the construction of parity-odd kurto spectra using composite-fields derived from filtered density fields to detect potential parity-violating signals in galaxy distributions. It details how these kurto spectra are formed through convolution integrals in Fourier space and explains their utility in canceling parity-even contributions, thus isolating any parity-violating signatures present in BOSS-DR12 and DESI-DR1 data sets.

Page 5: be affected by modelling uncertainties or systematics, we smooth the density field entering the Fourier-space composite fields, as described below. The CF two-point correlator is then defined as $\langle \delta_n(\mathbf{k}) \delta_m(\mathbf{k}') \rangle = (2\pi)^3 \delta_D(\mathbf{k} + \mathbf{k}') P_{\delta_n, \delta_m}(\mathbf{k})$, (2.3) where $\delta_D(\mathbf{k} + \mathbf{k}')$ is the Dirac-delta function, and P_{δ_n, δ_m} is the CF spectrum, which encodes the information of $(n + m)$ -point statistics. In pri

Page 6: Page 6 discusses the formulation and smoothing of composite-field spectra used to test parity with luminous red galaxies from BOSS and DESI surveys, introducing a weighted fluctuation field for survey data analysis that accounts for survey geometry and observational systematics through an FKP-like estimator. The smoothing process employs a Heaviside-step-function window to filter out scales smaller than approximately $7 h^{-1}$ Mpc.

Page 7: The page discusses the implementation of kurto spectra estimators within the `nbodykit` framework, detailing the use of configuration space and mesh techniques to measure two kurto spectra from BOSS-DR12 LRGs. It mentions the normalization process using FKP weights and describes the data division into NGC and SGC fields for analysis in the specified redshift range.

Page 8: Page 8 discusses the use of two mock catalogues—Patchy and Uchuu—in estimating data covariance and testing null hypotheses in galaxy surveys. The Patchy mocks use an approximate dark matter evolution model combined with stochastic biasing, while Uchuu mocks are based on N-body simulations and sub-halo abundance matching, each offering different advantages and potentially varying performance for higher-order statistics like the parity-odd kurto spectra examined here.

Page 9: The page discusses differences in weighting schemes between observed galaxy catalogs and mock suites (Patchy and Uchuu), highlighting how these variations impact the normalization factor estimations for BOSS data. It also notes that while the approximation methods using galaxy or random samples differ significantly in the trispectrum case, the paper opts to use the estimation from the galaxy catalogue. Additionally, it introduces the DESI-DR1 LRG sample used in their analysis, covering a redshift range of $0.4 < z < 1.1$ with specific number densities.

Page 10: Page 10 of "Testing parity with composite-field spectra of BOSS and DESI luminous red galaxies" details the DESI-DR1 weighting scheme, which uniformly applies across data and mocks. It also describes three sets of mock catalogues used for covariance estimation: EZmocks produced using an approximate fibre assignment algorithm, and two AbacusSummit N-body simulation sets with different fibre assignment methods to better match real data systematic effects. The page further outlines the normalization choices for power spectrum and trispectrum that extend BOSS methodology but use grid summation instead of Monte Carlo sampling.

Page 11: This page outlines the null hypothesis test for detecting parity-violating signals in luminous red galaxy spectra from DESI, focusing on comparing chi-square statistics between data and mock catalogs to identify significant deviations indicative of PV signals. The covariance matrix is numerically estimated from mocks and adjusted with a Wishart factor to account for non-Gaussianity due to the

limited number of mocks.

Page 12: Page 12 highlights the robustness and accuracy of numerical covariance estimates for kurto spectra over traditional 4PCF analyses due to lower data vector dimensions ($N_b = 70$ for BOSS-DR12, $N_b = 32$ for DESI-DR1) compared to the number of mocks. It further introduces a cross-chi-square statistic designed to mitigate data-mock mismatch issues, though it cautions that discrepancies in this statistic can arise from cosmological parity-violating signals, variance mismatches between data and mocks, or systematic correlated features not captured by the mocks.

Page 13: Page 13 of the paper emphasizes that the cross-chi-square ($\chi^2 \times$) is a data compression technique rather than a true chi-square statistic, used to detect cosmological parity-violation signals by comparing data and mocks with an arbitrary but consistent covariance. The page also introduces χ^2 null from Ref. [45], which highlights data-mock mismatches indicative of parity-even eight-point correlation differences, though no significant parity-violating signal was detected in the analyses performed.

Page 14: Figure 2 displays the kurto spectra $P2 \times 2$ and $P3 \times 1$ for BOSS-DR12 LRG samples in NGC and SGC, showing data measurements compared to Patchy and Uchuu mock averages. The χ^2 values of the observed kurto spectra do not indicate a non-zero parity-violating signal but suggest smaller null fluctuations in the BOSS data compared to Patchy mocks, aligning more closely with the higher-fidelity Uchuu mocks.

Page 15: Page 15 of "Testing parity with composite-field spectra of BOSS and DESI luminous red galaxies" presents Figure 3, illustrating χ^2 distributions for $P2 \times 2$ and $P3 \times 1$ kurto spectra from BOSS-DR12 using covariances from Patchy and Uchuu mocks. The figure compares data χ^2 values (solid red lines) to theoretical expectations (black curves), indicating varying degrees of consistency between the observed data and mock predictions across different models and spectral configurations.

Page 16: The page discusses χ^2 and $\chi^2 \times$ data compression results from BOSS-DR12 using Patchy and Uchuu mocks, showing that NGC-Full and SGC-Full data vectors are consistent with zero, indicating no shared features. The correlation matrices of both $P2 \times 2$ and $P3 \times 1$ kurto spectra are nearly diagonal, highlighting the advantages of kurto-spectrum approach in estimating covariance from mocks, which accounts for complex effects like non-Gaussianity and redshift-space distortions more effectively than full 4PCF analyses.

Page 17: The page discusses the χ^2 distributions between NGC and SGC for $P2 \times 2$ and $P3 \times 1$ measurements from BOSS-DR12 using covariances estimated from Patchy and Uchuu mocks. It highlights that the variance in Patchy is typically about 1.4 times larger than that of Uchuu, explaining why Uchuu realizations evaluated with Patchy covariance lie to the left of the expected χ^2 distribution, indicating statistic-dependent behavior between the two mock suites.

Page 18: Page 18 of the paper presents Figures 5 and 6, which compare correlation matrices for BOSS DR12 NGC using Patchy and Uchuu simulations, showing both raw correlations and relative differences, as well as the ratio of variances from these simulations for $P2 \times 2$ and $P3 \times 1$ spectra.

Page 19: On page 19, the core points are that DESI-DR1 measurements show smaller scatter compared to BOSS Patchy and Uchuu mocks due to larger survey volume and higher number density, consistent with theoretical predictions. The χ^2 distributions for DESI-DR1 $P2 \times 2$ and $P3 \times 1$ measurements do not detect a parity-violating signal, similar to the BOSS-DR12 findings, and the largest deviation from mocks is observed in the SGC-Full patch of $P2 \times 2$ at 2.4σ .

Page 20: Figures 7 and 8 present measurements of $P2 \times 2$ and $P3 \times 1$ respectively from DESI-DR1 data, with black dots representing data points and error bars derived from EZmocks; the blue solid line indicates the average of EZmocks, alongside a 1σ error shaded area for both NGC and SGC regions.

Page 21: Figure 9 presents the chi-square distributions for $P2 \times 2$ and $P3 \times 1$ measurements from DESI-DR1, showing comparisons between data and simulation models (EZMOCK, Abacus-FFA, Abacus-altMTL) with covariances estimated from EZMOCK. The figure highlights discrepancies indicated by sigma values, such as 0.9σ for NGC-Full $P2 \times 2$ and 3.1σ for NGC-Full $P3 \times 1$, demonstrating varying degrees of agreement between data and models.

Page 22: The page concludes that there is no significant parity violation (PV) signal detected in DESI-DR1 data across all patches except a marginal observation in the NGC-Full patch at 3.1σ , which does not constitute substantial evidence for PV. The AbacusSummit mocks show more scattered kurto spectra compared to EZMocks and are generally more consistent with the data than EZMocks,

suggesting that full N-body simulations better represent the data noise when no PV signal is present. The $\chi^2 \times$ statistics across different patch combinations confirm consistency with the null distribution from EZMocks without significant deviations exceeding 3σ , indicating that mock families behave similarly in the absence of a PV signal.

Page 23: Figure 10 presents the $\chi^2 \times$ statistics of $P2 \times 2$ measurements from DESI-DR1 using different covariances for NGC-Full and SGC-Full, showing various σ_{ez} values across different mocks and data comparisons, with black curves representing Gaussian distributions based on EZmocks.

Page 24: Page 24 presents Figure 11 and Figure 12. Figure 11 displays the χ^2 statistics for $P3 \times 1$ measurements from DESI using NGC-Full covariance, comparing data with simulations (EZmocks, Abacus-FFA, and Abacus-altMTL). Figure 12 shows correlation matrices for DESI DR1 NGC, illustrating differences in off-diagonal elements between EZMocks and the other two simulation methods, highlighting the variance in numerical covariance structures.

Page 25: The page concludes that no evidence for a parity-violating signal was found in BOSS-DR12 and DESI-DR1 data using kurto-spectrum estimators $P2 \times 2$ and $P3 \times 1$, with the covariance of these spectra being close to diagonal, indicating a noise-dominated regime. The analysis also shows that DESI-DR1 has significantly reduced scatter compared to BOSS-DR12, suggesting improved statistical sensitivity but still no detection of a cosmological parity-violating signal.

Page 26: Page 26 highlights that the kurto-spectra formalism offers practical advantages for testing parity violation by compressing trispectrum data more physically, but notes limitations due to incomplete DESI-DR1 data and restricted high-fidelity mocks. It also suggests improvements such as refining estimator normalization and weighting schemes, and extending the framework to build a bank of parity-odd templates for better model constraint identification with future larger datasets.

Page 27: The page discusses the calculation and comparison of $\chi^2 \times$ values for data and mock luminous red galaxy distributions from BOSS and DESI surveys, considering cross-patch correlations without assuming zero cross-covariance. The analysis shows that while $\chi^2 \times$ mock has a zero mean as expected, the data $\chi^2 \times$ can exhibit different distributions, influenced by factors such as k-cut choices and mock accuracy in representing data fluctuations.

Page 28: The page presents Figure 14, comparing data χ^2 with mock χ^2 -distributions for BOSS and DESI luminous red galaxies. For BOSS, the significance decreases with smaller scales, indicating mocks overestimate small-scale variance but remain within 3σ . In contrast, DESI shows irregular changes in significance with k_{max} , approaching 3σ levels but exhibiting unstable jumps around $k \sim 0.2 \text{ h Mpc}^{-1}$, leading to a conclusion that there is no significant non-zero signal and measurements are consistent with the null hypothesis.

Page 29: Page 29 of the paper lists several references that explore various aspects of parity violation in cosmological contexts, including theoretical frameworks for inflation, the impact of helical primordial magnetic fields, and methods to probe parity-violating modes through correlators and non-Gaussianities. These references collectively provide a comprehensive view of current research on detecting and understanding parity violations in large-scale structures and cosmic microwave background observations.

Page 30: Page 30 of the research paper discusses a study by Biagetti and Orlando, which explores the detection of primordial gravitational waves through galaxy intrinsic alignments, providing insights into potential observational methods for identifying these cosmic phenomena using large-scale structure data.

Page 31: The page lists recent studies and references related to detecting parity-violating signals in galaxy surveys, particularly focusing on BOSS and DESI datasets, as well as methods for analyzing large-scale structure data to measure non-Gaussianities and cosmic constraints using skew spectra. References also include foundational works on power-spectrum analysis and tools for large-scale structure simulations.

Page 32: The page lists several key references related to large-scale structure surveys and simulations, including the Uchuu-glam BOSS/eBOSS LRG lightcones, DESI Data Release 1, EZmocks for generating mock galaxy catalogs, and methods for addressing fiber assignment incompleteness in DESI data. Additionally, it cites works on high-fidelity N-body simulations like AbacusSummit and unbiased estimation techniques for model parameter confidences.

5. Probing the Solar ^8B Neutrino Fog with XENONnT

Focus: dark matter

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Summary: We report a 3.3σ measurement of coherent elastic neutrino-nucleus scattering from solar ^8B neutrinos using a 6.77 t exposure from the XENONnT experiment, inferring a solar ^8B neutrino flux of $(5_{-2}^{+3}) \cdot 10^6 \text{ cm}^{-2} \text{ s}^{-1}$, consistent with previous measurements. In the presence of the ^8B "neutrino fog", we find no evidence for light dark matter, and observe diminishing returns in sensitivity with increasing exposure. A 93% increase in exposure from the previous search improves the median sensitivity to $5 \text{ GeV}/c^2$ weakly interacting massive particles-nucleon cross section by 10%. The dataset was also used to measure the weak mixing angle at $0.02 \text{ GeV}/c$ momentum transfer and constrain physics beyond the Standard Model.

Page-by-page summaries

Page 1: The paper introduces the XENONnT experiment led by a collaborative team across multiple international institutions, aiming to detect and study solar ^8B neutrinos. The authors list their affiliations from various universities and research institutes in countries including the USA, Japan, Israel, Switzerland, France, Germany, Italy, China, Portugal, Sweden, and the United Arab Emirates.

Page 2: The XENONnT experiment reports a 3.3σ measurement of coherent elastic neutrino-nucleus scattering from solar ^8B neutrinos, inferring a flux consistent with previous measurements. The study also observes diminishing returns in sensitivity for light dark matter searches with increased exposure and uses the dataset to measure the weak mixing angle at low momentum transfer and constrain physics beyond the Standard Model.

Page 3: This page details adjustments and calibrations of the XENONnT detector, including changes in photoionization strength due to maintenance and filter installation, as well as updates on PMT performance and light/charge yield stability. It also discusses the radial correction for interaction position discrepancies and presents measurements of photon and electron gains (g_1 and g_2),

highlighting a reduction in S2 signals compared to previous runs.

Page 4: This page discusses the selection criteria and efficiencies for detecting neutrino-induced nuclear recoils in the XENONnT experiment. It highlights the energy ranges and signal requirements used to identify 8B solar neutrinos, detailing how S1 and S2 detection acceptances are evaluated and their uncertainties quantified. Additionally, it outlines light dark matter (LDM) and beyond-the-Standard-Model (BSM) neutrino scenarios considered for analysis, including modifications due to new vector mediators or a universal neutrino charge radius, while noting that the LDM contribution is nullified when constraining BSM interactions.

Page 5: The page details background suppression techniques for detecting 8B CEvNS events in XENONnT, emphasizing improvements in event reconstruction that reduce isolated S1 and S2 rates by about 20% in SR2. It describes the use of BDT classifiers on waveform features to discriminate between signal and AC backgrounds, achieving a significant reduction in expected AC events while validating the model with sideband datasets. Additionally, it outlines the statistical inference method using corrected S2 amplitude, pre-event correlations, and BDT scores to further refine background discrimination.

Page 6: The page details goodness-of-fit (GOF) tests applied to XENONnT data, including a conditional treatment of $S2_{pre}/\Delta t_{pre}$ based on p-value thresholds, which did not require exclusion after unblinding SR2 data with 25 events. The combined dataset of 62 events across all signal regions is found consistent with the expected 8B CEvNS signal plus background, leading to a four-dimensional binned likelihood analysis.

Page 7: Page 7 of the paper presents Fig. 4, showcasing upper limits on dark matter (DM) interactions at 90% CL, including SI DM-nucleon and SD DM-neutron scattering cross-sections, constraints on light mediators, and momentum-dependent MDDM interactions. The figure highlights improved sensitivity due to updated yields but also notes that increased exposure offers diminishing returns as experiments approach the "neutrino fog".

Page 8: Page 8 details the new upper limits on mediator coupling based on XENONnT data, showing significant rejection of background-only hypotheses and consistency with prior solar 8B neutrino flux measurements. The page also highlights the use of this dataset to probe beyond Standard Model (BSM) neutrino interactions and Low-Mass Dark Matter (LDM)–nucleon interactions, emphasizing the instrument's precision for low-energy neutrino studies and new physics searches.

Page 9: Figure 7 displays data and model projections for SR0, SR1, and SR2 regions, showing event counts with uncertainties alongside fitted components including the ^8B CEvNS signal (light green) and backgrounds from AC (purple), ER (blue), and neutrons (yellow).

Page 10: Page 10 of the research paper provides a comprehensive list of references detailing various aspects and advancements in the XENONnT experiment, including detector improvements, background reduction techniques, and theoretical frameworks for probing dark matter and neutrino interactions. References span from foundational works to recent contributions that enhance the sensitivity and reliability of the XENONnT apparatus.

Page 11: Page 11 lists references related to dark matter searches and coherent elastic neutrino-nucleus scattering (CEvNS) experiments, including recent studies by PandaX-4T, LZ, DarkSide-50, and CRESST-II. Additionally, it cites theoretical work on nuclear structure calculations relevant for weakly interacting massive particle (WIMP) scattering with chiral effective field theory currents.

6. Unveiling axion signals in galactic supernovae with future MeV telescopes

Focus: dark matter

Authors: Zhen Xie, Jiahao Liu, Bing Liu, Ruizhi Yang

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Summary: Axion-like particles (ALPs) produced via the Primakoff process in the cores of Galactic core-collapse supernovae (SNe) could convert into MeV-energy gamma-rays through interactions with the Milky Way's magnetic field. To evaluate the detection prospects for such signals, we perform sensitivity projections for next-generation MeV telescopes by combining hypothetical instrument responses with realistic background estimates. Our analysis incorporates detailed simulations of the expected ALP flux from nearby SNe, the energy-dependent conversion probability in Galactic magnetic fields, and the telescope's angular/energy resolution based on advanced detector designs. Background components are modeled using data from current MeV missions and extrapolated to future sensitivity regimes. Our simulations demonstrate that next-generation telescopes with improved effective areas and energy resolution could achieve sensitivity to photon-ALP couplings as low as $g_{\gamma\gamma} \approx 1.61 \times 10^{-13} \text{ GeV}^{-1}$ for ALP masses $m_a < 10^{-9} \text{ eV}$ in Galactic Center. These results indicate that future MeV missions will probe unexplored regions of ALP parameter space, with conservative estimates suggesting they could constrain $g_{\gamma\gamma}$ values two orders of magnitude below current astrophysical limits. Such observations would provide the most stringent tests to date for axion-like particles as a dark matter candidate in the ultra-light mass regime.

Page-by-page summaries

Page 1: The research evaluates the potential of future MeV telescopes to detect axion-like particle (ALP) signals from Galactic core-collapse supernovae by simulating ALP conversion into MeV-energy γ -rays in the Milky Way's magnetic field. The study projects that next-generation telescopes could achieve sensitivity down to $g_{\gamma\gamma} \approx 1.61 \times 10^{-13} \text{ GeV}^{-1}$ for ALP masses $m_a \lesssim 10^{-9} \text{ eV}$, probing previously unexplored regions of ALP parameter space and providing stringent tests for ultra-light dark matter candidates.

Page 2: Axion-like particles (ALPs) can be generated in supernova cores through the Primakoff process, converting thermal photons into ALPs via interactions with electrostatic fields of charged particles. This mechanism is crucial for modeling photon-ALP conversions and detecting ALPs using next-generation MeV telescopes like e-ASTROGAM and AMEGO-X, which are designed to capture high-energy photon signals from astrophysical sources such as supernovae.

Page 3: The page details the calculation of axion (ALP) production rates within supernova cores using the Primakoff process, providing a differential rate equation that depends on photon energy, core temperature, and effective coupling constants. It also presents an integrated ALP flux estimation method, critical for understanding high-energy astrophysical signals and dark matter implications, along with a simplified fitting expression commonly used in SN neutrino studies.

Page 4: The page discusses how future MeV telescopes with assumed effective areas of 100 cm^2 and a point spread function of 2° will enhance sensitivity in detecting axion-induced gamma-ray signals from supernovae. It highlights Betelgeuse as a prime candidate for such studies due to its imminent potential for a SN explosion, offering critical insights into massive star evolution and the properties of both ultralight and massive axions through observable phenomena like enhanced cooling rates and distinctive gamma-ray spectra.

Page 5: This page discusses the use of M31 (Andromeda Galaxy) as a valuable target for studying supernovae and their potential axion signals, alongside Betelgeuse, SN 1987A, and the Galactic Center. It details the background gamma-ray intensity calculations in these regions using data from INTEGRAL/SPI below 10 MeV and Fermi-LAT above 50 MeV, with Fig. 2 illustrating the interpolated background flux estimates for different sky positions relevant to supernova observations.

Page 6: The page details the selection of an energy range (1–100 MeV) and a 10-second exposure time to estimate background counts, yielding a conservative 95% confidence upper limit of 4.14 counts for SN signals. The study also evaluates ALP parameter constraints, with sensitivities reaching up to $1.61 \times 10^{-13} \text{ GeV}^{-1}$ in the Galactic Center and $1.26 \times 10^{-13} \text{ GeV}^{-1}$ for Betelgeuse at masses $m_a \lesssim 10^{-9} \text{ eV}$, underlining the potential of future MeV detectors despite current limitations.

Page 7: The page emphasizes that future MeV detectors can impose stronger constraints on axion-like particle (ALP) parameters during galactic supernova explosions by leveraging early neutrino signals as

precursors. It highlights that while increasing detector size may not significantly boost observation probability due to the rarity of SN events, utilizing neutrinos for timing information is key to optimizing detection sensitivity and improving ALP parameter constraints.

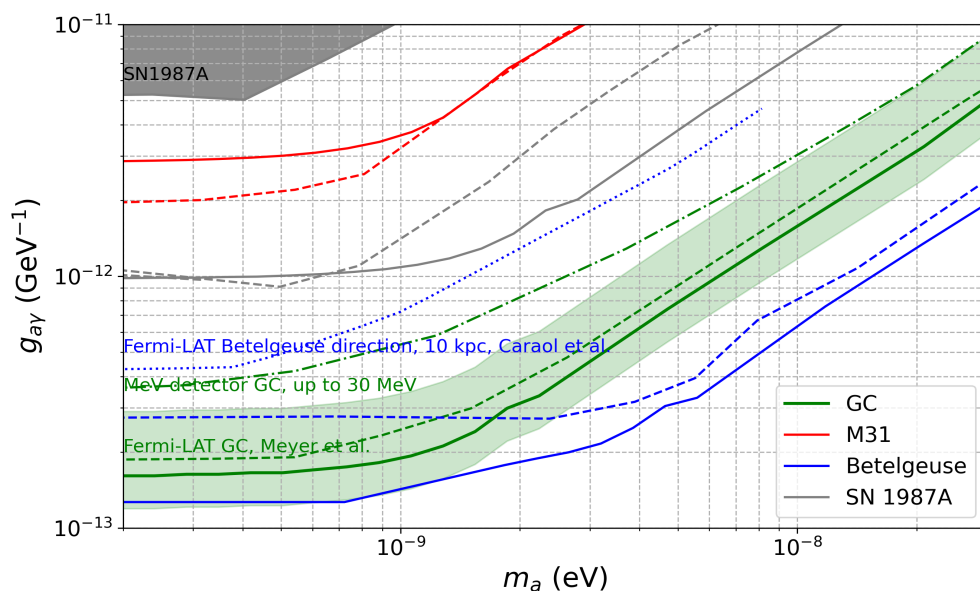
Page 8: Page 8 discusses the capabilities of neutrino detectors like SK, SNO, and upcoming facilities such as Hyper-Kamiokande and DUNE, which can detect up to approximately 10^4 neutrino events from an SN explosion at 10 kpc. These detectors provide early alerts for supernova explosions, offering critical temporal information that enhances the understanding of SN dynamics and serves as a benchmark for data analysis in future MeV missions investigating axion-like particle properties.

Page 9: The page lists numerous references that detail studies on axion physics and detection methods, including theoretical works on axions in supernova gamma-ray spectra and experimental constraints from various telescopes and missions such as the Fermi Large Area Telescope and future MeV telescopes like e-Astrogam and AMEGO-X.

Page 10: This page lists references discussing heavy axion-like particles in core-collapse supernovae, multimessenger signals from such phenomena, and the potential detection of these signals with future MeV telescopes, emphasizing recent research on shock revival and γ -ray emissions. Additionally, it cites foundational astrophysical resources relevant to understanding galactic observations and particle physics in stellar environments.

Page 11: The page lists references discussing neutrino signals from supernovae, their potential to identify progenitor stars and explore dark photon parameter space, and highlights advancements in neutrino observatories such as KamLAND, Super-Kamiokande, IceCube, Hyper-Kamiokande, JUNO, and DUNE.

Selected figures



7. PhageBench: Can LLMs Understand Raw Bacteriophage Genomes?

Focus: dark matter

Authors: Yusen Hou, Weicai Long, Haitao Hu, Houcheng Su, Junning Feng, Yanlin Zhang

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Summary: Bacteriophages, often referred to as the dark matter of the biosphere, play a critical role in regulating microbial ecosystems and in antibiotic alternatives. Thus, accurate interpretation of their genomes holds significant scientific and practical value. While general-purpose Large Language Models (LLMs) excel at understanding biological texts, their ability to directly interpret raw nucleotide sequences and perform biological reasoning remains underexplored. To address this, we introduce PhageBench, the first benchmark designed to evaluate phage genome understanding by mirroring the workflow of bioinformatics experts. The dataset contains 5,600 high-quality samples covering five core tasks across three stages: Screening, Quality Control, and Phenotype Annotation. Our evaluation of eight LLMs reveals that general-purpose reasoning models significantly outperform random baselines in phage contig identification and host prediction, demonstrating promising potential for genomic understanding. However, they exhibit significant limitations in complex reasoning tasks involving long-range dependencies and fine-grained functional localization. These findings highlight the necessity of developing next-generation models with enhanced reasoning capabilities for biological sequences.

Page-by-page summaries

Page 1: PhageBench introduces a benchmark to evaluate general-purpose Large Language Models (LLMs) on interpreting raw bacteriophage genomes across five core tasks, revealing their potential but also significant limitations in complex biological reasoning. The study highlights the need for next-generation models with enhanced capabilities to understand biological sequences effectively.

Page 2: PhageBench introduces raw phage genome annotation as a challenging testbed for evaluating the reasoning capabilities of general-purpose LLMs across three critical stages: Screening, Quality Control, and Phenotype Annotation. The benchmark consists of over 5,600 samples and evaluates eight advanced LLMs, revealing their potential in phage identification and host prediction tasks but also highlighting limitations in handling long-range dependencies for complex syntactic analysis without domain-specific pre-training.

Page 3: PhageBench introduces a comprehensive evaluation dataset focusing on the direct understanding and reasoning of raw bacteriophage genomic sequences using general-purpose LLMs, marking the first study to assess such models on authentic whole-genome analysis tasks. The dataset integrates task-specific processing rules and standardized formats to rigorously evaluate LLM performance in biological contexts.

Page 4: PhageBench evaluates LLMs on five core tasks of phage genome analysis—phage identification, contamination detection, completeness estimation, lifestyle classification, and host prediction—using rigorously processed sequences from authoritative databases to assess the models' ability to discern phage signatures, evaluate sequence quality, and infer biological functions.

Page 5: Page 5 of "PhageBench: Can LLMs Understand Raw Bacteriophage Genomes?" details the dataset's statistical properties and the evaluation setup for assessing LLMs on phage genome understanding. The PhageBench dataset, comprising 5,600 samples across five tasks with diverse genomic sequence characteristics, is used to evaluate eight advanced LLMs under zero-shot conditions, focusing on their inherent biological reasoning capabilities using classification accuracy as the primary metric.

Page 6: Table 1 highlights that reasoning-enhanced models, such as Gemini-3-flash and GPT-5.2, significantly outperform non-reasoning models in understanding raw bacteriophage genomes, achieving accuracies of up to 56.33% and 50.95%, respectively. The study also observes that while models excel at recognizing statistical genomic features like phage contig identification (up to 70% accuracy) and host prediction (62.50%), they face challenges in tasks requiring global structural assessment or fine-grained inference, such as contamination detection and lifestyle classification.

Page 7: The page presents a comparison of model performance in bacteriophage genome tasks, showing that while some models like GPT-5.2 and Gemini-3-flash exhibit significant improvements with Chain-of-Thought (CoT) prompting across most tasks, others show varied responses, particularly in

tasks requiring long-range dependencies or precise retrieval of functional markers. Additionally, the impact of test-time scaling is analyzed, demonstrating that CoT generally enhances performance but can lead to regressions in specific cases, indicating a nuanced application for biological data.

Page 8: Page 8 of "PhageBench: Can LLMs Understand Raw Bacteriophage Genomes?" discusses the model's limitations in grounding abstract biological knowledge with precise sequence data, highlighting frequent errors related to incorrect gene presence claims. The page also proposes strategies for future research, including multi-modal alignment and tool-augmented reasoning, to enhance models' ability to interpret raw nucleotide sequences accurately. Additionally, it suggests optimizing global dependency modeling to improve the correlation of distal features in long sequences.

Page 9: Page 9 outlines the limitations of PhageBench, including data scope restrictions that exclude larger phage entities, suboptimal tokenization strategies for genomic data using Byte Pair Encoding (BPE), and risks of data contamination with closed-source models. The page also addresses ethical considerations, confirming adherence to biosafety regulations and urging responsible AI development in computational biology.

Page 10: Page 10 of the paper "PhageBench: Can LLMs Understand Raw Bacteriophage Genomes?" lists numerous references related to bacteriophage genomics, language models applied to genomics, and various tools and benchmarks for genomic sequence analysis. These citations provide context for existing research and methodologies that are pertinent to assessing large language models' (LLMs) ability to understand raw bacteriophage genomes.

Page 11: Page 11 discusses the broader impact of PhageBench, highlighting its potential to advance both AI and computational biology by extending Large Language Models' capabilities from processing scientific literature to decoding genomic data. It underscores the significance of efficient genome annotation for addressing antibiotic resistance through phage therapy and advocates for interdisciplinary collaboration to bridge AI and biological research.

Page 12: Page 12 details the construction of PhageBench's dataset, including high-quality phage genome sourcing from PhageScope and NCBI RefSeq, along with stringent preprocessing to ensure sequence quality. It also describes the balanced binary classification dataset for phage contig identification, and a mock data generation process simulating biological contamination at specific ratios using three insertion strategies: Prefix, Suffix, and Internal.

Page 13: Page 13 details the construction of PhageBench's completeness estimation and phenotype annotation tasks, including lifestyle classification and host prediction, with specific instructions for each task listed in Table 3. The dataset ensures robust statistical properties by controlling sequence lengths and GC content variability while employing a hardness-aware distractor generation strategy to challenge models' biological understanding through multiple-choice questions.

Page 14: Figures 7 to 9 depict detailed class distributions and balancing strategies across various tasks, ensuring statistical fairness through strict ratio maintenance and uniform distribution. For instance, phage identification maintains a strict 1:1 positive-to-negative ratio with negative samples stratified into six biological categories, while contamination detection and completeness estimation tasks feature balanced datasets and quality tiers, respectively. The answer key distribution across all tasks is uniformly randomized to mitigate position bias, with correct options (A, B, C, D) evenly distributed.

Page 15: Page 15 of "PhageBench: Can LLMs Understand Raw Bacteriophage Genomes?" details the standardized input-output format used in evaluating LLM performance on genomic tasks, providing examples for various classifications such as phage contig identification and host prediction. The evaluation section outlines implementation specifics like API access, temperature settings, reasoning limits, and baseline comparisons to gauge model accuracy effectively.

Page 16: The page describes two prompt variations for evaluating LLM performance on genomic questions: a Zero-shot CoT prompt that requires step-by-step reasoning and an analysis field, and a standard Zero-shot prompt without these features. It also discusses model limitations in distinguishing phages from plasmids due to reliance on broad statistical features rather than specific genetic structures, and their difficulty in detecting low-level contamination (12.5%) due to similar statistical properties of host and phage sequences.

Page 17: The page discusses qualitative results of how LLMs interpret raw phage genome sequences, focusing on success cases where Gemini-3-flash effectively identifies phage contigs by recognizing high gene density and modular organization typical of Siphoviridae or Myoviridae families, and accurately predicts host families through statistical genomic signatures.

Page 18: The page details the performance and limitations of LLMs on PhageBench tasks, highlighting their ability to correctly identify high GC content associated with Pseudomonas genus but also their failure in handling heterogeneous sequences and tendency towards structural and knowledge-driven hallucinations.

Page 19: The analysis identifies characteristic bacteriophage genomic features in the sequence, including high gene density, modular organization, and specific regulatory elements, confirming its origin from a bacteriophage genome (B). The sequence also contains genes encoding structural proteins and DNA replication machinery typical of Siphoviridae or Myoviridae families.

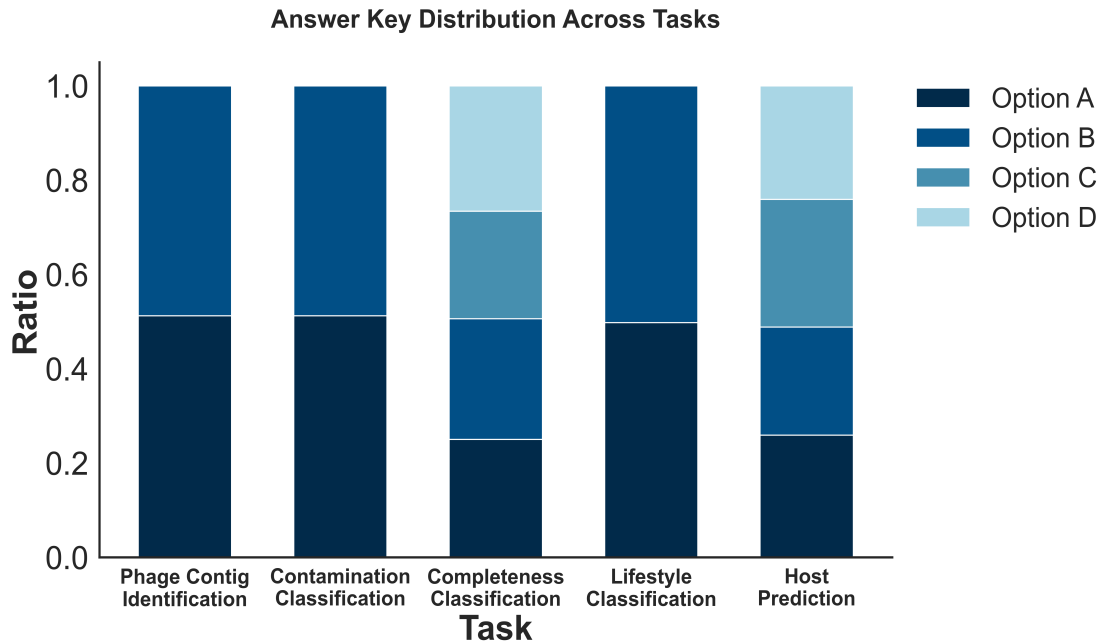
Page 20: The analysis reveals numerous genes encoding characteristic phage structural proteins and machinery, with no significant regions of non-phage host genomic DNA identified. However, the ground truth indicates a contamination ratio of 50%, suggesting the model failed to detect host contamination effectively in this case study.

Page 21: The page describes the process of estimating genome completeness for a bacteriophage sequence, involving the identification of direct terminal repeats (DTRs), essential genes like capsid and tail proteins, and the formulation of a scoring system to categorize the genome. Despite identifying characteristic phage genes and structural features indicative of a complete genome, the estimated category "Complete" does not match the ground truth labeled as "High-quality."

Page 22: The analysis identifies the provided genome sequence as closely resembling that of PhiX174, a virulent bacteriophage with a strictly lytic lifestyle, thereby concluding that answer (B) is correct. However, the ground truth information labels the phage as temperate (lysogenic-capable), highlighting a discrepancy and indicating a failure in lifestyle classification by the model.

Page 23: The analysis of the bacteriophage sequence reveals a high GC content (approximately 60-65%), characteristic of the Pseudomonadaceae family, alongside genomic features consistent with phages infecting Pseudomonas species. The correct prediction for the host family is (A) Pseudomonadaceae.

Selected figures



8. Mirror Dual Symmetry in Physics

Focus: dark matter

Authors: Lucas Lamata

Published: 2026-04-07T11:43:44+00:00

Summary: The quantum Rabi model has been a useful and pedagogical quantum model in the past decades, sufficiently simple to be solved analytically and intuitively understood, while sufficiently complex as to provide highly non-trivial eigenstates and a practical description of quantum optical platforms for quantum technologies. The Dirac equation, especially when restricted to 1+1 dimensions, is a simple toy model as well, but its easy diagonalization enabled historically to connect the electron spin to the fermionic statistics, among others. Both models share a symmetry at the purely mathematical level, namely, the spectra of each one has a dual equivalent under energy sign change, that I name a mirror dual symmetry. Usually, one quantizes these equations by assuming a ground state energy for the bosonic mode. But there is another option for the interpretation of the Hamiltonian, as I will argue, that is to assume a total symmetry principle, namely, that the total energy is zero at all times, for either the quantum Rabi model or the Dirac equation, and impose the constraint that every positive energy excitation has a mirror excitation of negative energy. This possibility, which was, apparently, ignored in the times when Paul Dirac was studying the implications of his equation, would avoid the worries in the scientific community that the negative energy solutions would decay until minus infinity, thus obviating the necessity to build a highly artificial Dirac sea, and instead impose what has always been successful in Physics, which is the enforcement of symmetry principles. Assuming a total symmetry principle, many of the problems of current Physics, such as renormalization of quantum gravity, dark matter, and dark energy, may possibly be automatically solved. One obvious result would be the automatic cancellation of the zero point energy.

Page-by-page summaries

Page 1: The page introduces the "total symmetry principle," positing that the Universe, considered globally, has no structure or dimensions and can be viewed as a Hilbert space of dimension 1 or a metric space of dimension 0. This principle suggests that every positive energy excitation in systems like the quantum Rabi model or Dirac equation has a mirror negative energy excitation, leading to zero total energy, which could resolve issues such as renormalization and zero point energy discrepancies.

Page 2: Page 2 elaborates on the concept of mirror dual symmetry, where each quantum excitation in our Universe has a dual excitation with negative energy, momentum, and charges in a CPT-symmetric "mirror" Universe. This symmetry is illustrated through modifications to the quantum Rabi and Dirac equations, and its implications are further explored in contexts such as black hole radiation and holographic projections, suggesting that dark matter and dark energy could be residual effects of entanglement with this mirror dual Universe.

Page 3: Page 3 of "Mirror Dual Symmetry in Physics" lists references that underpin the theoretical frameworks discussed, ranging from foundational works on quantum theory and gravity to more contemporary explorations into CPT symmetry and the origins of gravitational laws, highlighting the interdisciplinary nature of the research.

9. Stellar Population Characterisations in nearby, dusty Early-Type Galaxies

Focus: galaxies

Authors: Ron W. Savage, Anne E. Sansom, David H. W. Glass

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Summary: Dust in Early-Type galaxies (ETGs) may originate from internal or external sources. In this paper we study the stellar populations of particularly dusty ETGs to search for evidence of the dust's origin. Using the Southern African Large Telescope (SALT), we obtained long-slit optical spectra within the effective radius (R_e), along the major axis of 15 nearby ETGs, selected from the GAMA and Herschel-ATLAS surveys for their high levels of interstellar dust. Using full spectrum fitting and Lick index fitting we analysed their major axis kinematics and stellar population characteristics. We used stellar population models from the newly developed sMILES library and from the empirical MILES library. Kinematic results show that most of our sample of dusty ETGs are rotationally supported and there are no detectable kinematic discontinuities. 12 of our sample of 15 dusty ETGs show evidence of young/intermediate age stellar population components suggesting ongoing/recent star formation. Using simulations, we show that these recent (1–Gyr) populations are not artefacts of the fitting process or data. As a check with a control sample we use stacked SDSS spectra and find that dusty ETGs show a component with intermediate age, whereas non-dusty ETGs do not. Age, metallicity and α -element abundance ratio increase with increasing central velocity dispersion in the SALT spectra, as seen in previous studies of ETGs, but with larger scatter in our sample. Given our stellar population findings, we discuss formation scenarios that might cause or rule out a high dust/molecular gas content.

Page-by-page summaries

Page 1: The page introduces research on dusty Early-Type Galaxies (ETGs), highlighting that these galaxies, despite typically having low star formation rates and older stellar populations, can contain significant dust levels. The study uses optical spectra from the Southern African Large Telescope (SALT) to analyze 15 nearby ETGs selected for their high dust content, aiming to identify whether the dust originates internally or externally by examining stellar population characteristics and kinematics.

Page 2: The page discusses the uncertainty around dust sources in early-type galaxies (ETGs), suggesting contributions from both internal and external sources, with evidence pointing to variability that cannot be solely attributed to continuous stellar mass loss. It highlights the need to investigate stellar population properties to understand evolutionary paths influenced by interactions or mergers, while noting that detailed star formation histories relative to cold ISM content have not been extensively studied. The text also touches on how absorption features in optical spectra can provide information about stellar populations but acknowledges the complexity of extracting parameters like age, metallicity, and $[M/Fe]$ due to parameter degeneracies.

Page 3: The page details the construction of an initial complete sample of nearby, dusty early-type galaxies (ETGs) from GAMA data with specific redshift and morphological criteria, resulting in a parent sample of 4458 galaxies. Strong AGN were excluded to avoid contamination of submillimeter spectra, leading to a final selection of 202 dusty ETGs, among which 15 were further observed for their stellar populations using SALT and IRAM 30m telescopes. These selected dusty ETGs exhibit a range of stellar masses from 10^{10} to $10^{11.3} M_\odot$ and dust mass ratios (DMR) from 7.3×10^{-5} to 1.3×10^{-3} .

Page 4: Page 4 details the physical properties and morphological classification of 15 target dusty early-type galaxies (ETGs), including stellar mass, dust mass, half-light radius, ellipticity, redshift, and visual morphology. The page also outlines the data reduction process for SALT RSS spectra, emphasizing corrections for scattered light to mitigate systematic biases in stellar population fitting and velocity dispersion calculations.

Page 5: Table 2 summarizes longslit observations of GAMA galaxies using SALT/RSS, detailing observation dates, exposure times, airmass values, slit angles, and the number/S/N ratio of spatial bins for spectral extraction. The paper describes methodologies to extract stellar kinematics and population parameters via full spectrum fitting with pPXF, employing MILES and sMILES SSP libraries while noting that α -element enhancement estimation required Lick index fitting due to insufficient S/N for radial profiles.

Page 6: Page 6 of the paper "Stellar Population Characterisations in nearby, dusty Early-Type Galaxies" discusses the kinematic analysis of 15 target early-type galaxies using pPXF with MILES and

sMILES SSPs. The results show that central velocity dispersion (σ_0) values are consistent within 1-sigma uncertainties across different fitting methods, indicating a typical range for intermediate mass ETGs of ≈ 100 to 200 km s^{-1} . Additionally, the paper presents kinematic parameters including recession velocity and velocity dispersion for these galaxies from central aperture spectra.

Page 7: The page details kinematic profiles and stellar population analyses of nearby, dusty early-type galaxies (ETGs), using MILES and sMILES SSP fittings. It shows significant rotational support in these galaxies without kinematic discontinuities but with moderate S/N ratios. Age and metallicity estimates are obtained via pPXF fitting with linear regularization, revealing a mix of old, intermediate-age, and young stellar populations across the 15 galaxy sample. No clear trends in star formation history (SFH) related to group membership are observed.

Page 8: Page 8 of "Stellar Population Characterisations in nearby, dusty Early-Type Galaxies" presents Figure 5, illustrating pPXF fitting outputs for two galaxies (GAMA422436 and GAMA99687), showing distinct differences in their stellar populations. GAMA422436 has a single old, metal-rich population ($>10 \text{ Gyr}$), while GAMA99687 contains both an older ($>\approx 10 \text{ Gyr}$) and younger ($\approx 0.9 \text{ Gyr}$) population at sub-solar metallicities, highlighting the diversity in stellar populations among early-type galaxies.

Page 9: Page 9 discusses the central aperture Age $_{\text{pPXF}}$ and $[M/H]_{\text{pPXF}}$ results for dusty early-type galaxies (ETGs) plotted against σ_0 , showing broad comparability between MILES and sMILES SSPs and a loose trend of increasing Age $_{\text{pPXF}}$ and $[M/H]_{\text{pPXF}}$ with increasing σ_0 . The page also notes that pPXF full spectrum fittings using sMILES 3D grids reveal a decreasing trend of $[Fe/H]_{\text{pPXF}}$ with increasing σ_0 , attributed to the use of α -element dependent SSPs which do not account for variations in individual α -elements.

Page 10: Table 4 on page 10 presents the results of age and metallicity determinations from pPXF full spectrum fitting and Lick index SSP fitting for central apertures in several galaxies, showing discrepancies between different fitting methods and highlighting uncertainties calculated via Monte Carlo simulations.

Page 11: The page details the use of Lick indices for fitting single stellar population (SSP) models to extract average age, metallicity, and α -element abundance from central aperture spectra of dusty early-type galaxies. The authors identify systematic differences between their results and those from full spectrum fitting methods, attributing these discrepancies to biases in Lick index SSP fits towards younger ages and higher metallicities in composite stellar populations. Comparisons with passive ETGs suggest that the observed trends are more pronounced due to the presence of young stellar components in dusty galaxies.

Page 12: The page presents a comparison between dusty and non-dusty early-type galaxies (ETGs), revealing that the dusty ETG stack is approximately $2.8 \pm 0.6 \text{ Gyr}$ younger than the non-dusty stack, with no significant difference in mean metallicity or $[Fe/H]$ ratio. Additionally, pPXF analysis shows evidence of an intermediate-age stellar population contribution in dusty ETGs, which is absent in non-dusty ETGs.

Page 13: Page 13 discusses diagnostic plots and evolutionary scenarios for understanding the origins of dust in dusty early-type galaxies (ETGs). The page highlights that intermediate-age stellar populations (≈ 0.5 to $\approx 2 \text{ Gyr}$) are often observed, suggesting dust could be associated with recent gas accretion events such as minor mergers. These scenarios are explored through age vs. metallicity grids and compared against different merger mass ratios, indicating possible external accretion of gas and dust in these galaxies.

Page 14: Figure 8 compares Lick index SSP fits for SDSS stacks of non-dusty and dusty early-type galaxies, illustrating confidence levels and chi-squared contours. Figure 9 shows luminosity-weighted pPXF fits and alternative scalings for these spectra, revealing an intermediate-age stellar population present in dusty ETGs but absent in non-dusty ETGs.

Page 15: Page 15 of the paper presents Table 5, detailing mass ratios and metallicities for minor merger scenarios in nearby, dusty early-type galaxies (ETGs). It also discusses how the stellar populations of these galaxies show more intermediate-age contributions compared to other ETG samples not selected by dust content. Specifically, GAMA272990 exhibits evidence of a recent gas-rich minor merger, indicated by its molecular gas disc and rejuvenated star formation, suggesting a connection between the merger event and the youngest stellar population component.

Page 16: The page discusses that three out of fifteen observed dusty early-type galaxies (ETGs) show only old stellar populations despite their high dust content, which is unexpected. Analysis using pPXF age vs. metallicity grids suggests that lower metallicities in intermediate-age stellar components indicate some dust may be externally sourced, possibly through minor mergers. Future high S/N spectroscopic studies are necessary to clarify these findings.

Page 17: Stellar Populations in nearby, dusty ETGs 17 Goudfrooij P., de Jong T., 1995, A&A, 298, 784 Griffith E., Martini P., Conroy C., 2019, MNRAS, 484, 562 Holwerda B. W., et al., 2024, Publ. Astron. Soc. Australia, 41, e115 Jegatheesan K., Johnston E. J., Häußler B., Nedkova K. V., 2024, A&A, 684, A32 Jegatheesan K., Johnston E. J., Häußler B., Lassen A. E., Riffel R., Chies- Santos A. L., 2025, A&A, 6

Page 18: The page presents $[M/H]$ versus $\log_{10}$ Age grids for ETGs observed with SALT and fitted using MILES SSPs, revealing diverse stellar populations ranging from old, high metallicity to younger, lower metallicity components. Some galaxies show evidence of continuous star formation or contain anomalous results due to strong emission lines interfering with full spectrum fitting.

Page 19: Figure A2 on page 19 of the paper presents $[M/H]$ vs. $\log_{10}$ Age grids from pPXF luminosity-weighted full spectrum fitting of synthetic test spectra using MILES SSPs, illustrating various combinations of metal-rich and metal-poor stellar populations with different age components, including continuous star formation models.

Selected figures



10. The Broken Similarity: Sinking and Merging of Dark Matter Subhalos Across Hierarchical Levels

Focus: dark matter

Authors: Wenkang Jiang, Jiaxin Han, Kun Xu, Victor J. Forouhar Moreno, Feihong He, Zhaozhou Li, Chunyan Jiang, Yipeng Jing, Xiaohu Yang

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Summary: We investigate hierarchical mergers among subhalos within a Λ CDM simulation using the subhalo finder. Unlike previous methods, tracks subhalo evolution across hierarchy levels, identifying the coalescence of subhalo cores in phase-space as a "sinking" event. This coalescence marks a distinct stalled phase in orbital decay, providing a physically motivated and natural definition of a resolved merger. Our main findings include: 1) Over 90% of sinking events occur between adjacent subhalo levels, while cross-level pathways arise from tidal stripping, group infall, and numerical constraints. 2) Resolved mergers are predominantly \ major mergers (mass ratios $> 1:10$), while the occurrence of minor mergers decreases with the dynamical age of the host halo. 3) Although deep-level subhalos have low mass ratios relative to the host halo, their high mass ratios relative to direct parents significantly boost merger statistics. Consequently, the satellite-satellite merger rate can rival or exceed the central-satellite rate at lower mass thresholds. 4) Satellite-satellite mergers are spatially biased toward the outer regions of the host, suggesting that the central tidal field suppresses their orbital decay. 5) A bidirectional sinking detection recovers 32% more sinking events than the original algorithm, revealing that child-dispersion-driven mergers are dominated by tidal heating at the final stage of sinking, while parent-dispersion-driven and doubly-identified events proceed primarily via orbital decay. Altogether, these results reveal a complex landscape of hierarchical satellite mergers which deviate from the self-similarity of host halo mergers, due to additional physical processes including dynamical friction and the scale-dependent halo growth history.

Page-by-page summaries

Page 1: This page introduces a study using the hbt+ subhalo finder to investigate hierarchical mergers among dark matter subhalos within a Λ CDM simulation, highlighting that over 90% of sinking events occur between adjacent levels and that resolved mergers are predominantly major mergers. The study also reveals spatial biases in satellite–satellite mergers towards the outer regions of host halos due to tidal suppression and shows that bidirectional sinking detection increases identified merging events by 32%, uncovering different mechanisms driving these mergers.

Page 2: The page discusses the critical role of subhalo mergers in galaxy evolution within the Λ CDM paradigm, emphasizing the importance of the merger timescale and rate, which are influenced by factors such as infall redshift, orbital parameters, and mass ratios. It highlights the complexities arising from varied definitions of what constitutes a subhalo merger and the hierarchical nature of accretion, leading to both central-satellite and satellite-satellite mergers. The authors use a dark-matter-only cosmological simulation with the hbt+ subhalo finder to investigate these mergers through a physically motivated criterion they term "subhalo sinking."

Page 3: The paper details the complexities in subhalo mergers that deviate from self-similarity due to additional physical processes post-accretion. Section 2 introduces the simulation parameters and the hbt+ algorithm used for tracking subhalos, defining hierarchical levels, and identifying sinking subhalos through a phase-space distance criterion.

Page 4: The page details an enhanced algorithm for detecting subhalo sinking events by evaluating phase-space distances bidirectionally, capturing both the standard satellite-to-parent and parent-to-satellite dynamics. This refined approach identifies approximately 32% more sinking events in massive halos compared to the original single-sided criterion, though these additional events involve rapid disruption due to tidal heating. Figure 1 illustrates a representative sinking event's two-phase evolution: an initial rapid inspiral followed by a stalling phase marked by oscillations within the core scales of the central subhalo.

Page 5: The page discusses criteria for identifying stable sinking events of dark matter subhalos, applying cuts on peak mass ($m_{\text{peak}} \geq 100 m_p$) and minimum sinking mass ($m_{\text{sink}} \geq 20 m_p$). It identifies 415,976 sinking events in host halos with masses $[10^{12}, 10^{13}] M_{\odot} h^{-1}$ and 68,997 in hosts of mass $[10^{13}, 10^{14}] M_{\odot} h^{-1}$. The page also introduces a refined classification based on hierarchical levels to capture the detailed sinking pathways of subhalos, revealing that most sinking events occur between a subhalo and its direct parent but with a non-negligible fraction involving cross-level sinks.

Page 6: Figure 3 illustrates the distribution of final coalesced parent subhalos for child subhalos across different levels and host halo masses. Figure 4 shows that Orphaned Sinking and Overtaking Sinking are the dominant pathways in cross-level sinking events, indicating significant roles of both tidal stripping by massive nearby subhalos and insufficient particle resolution in these processes. The findings have implications for semi-analytical models regarding merger timescales post-accretion.

Page 7: The page discusses the sinking rates of dark matter subhalos within hierarchical levels, introducing group-level and halo-level definitions to measure these rates more precisely. The group-level sinking rate quantifies mergers between child subhalos and their direct parents within specific mass ranges, while the halo-level sinking rate aggregates all such events within an entire host halo. Key results show that subhalo sinking predominantly involves massive child subhalos with strong dynamical friction effects, and level-2 sinking rates are significantly lower than level-1 rates at $z = 0$ due to factors like starvation and tidal mass loss.

Page 8: The page discusses sinking rates of dark matter subhalos across different hierarchical levels and redshifts, showing that tidal heating can delay orbital decay and suppress merger rates. It highlights a substantial population of low-mass-ratio sinking events in level-2 subhalos due to convolution with the peak mass function of level-1 subhalos, and notes a shift towards lower peak mass ratios for sunken subhalos at higher redshifts and host halo masses.

Page 9: The page discusses the redshift evolution of subhalo sinking rates, noting that lower mass ratio subhalos sink earlier in recently-formed halos, leading to a systematic shift towards lower merger ratios at higher redshifts. It also highlights a discrepancy with previous work by A. R. Wetzel et al., which underestimates satellite–satellite mergers due to different sample selection criteria and normalization methods. The authors emphasize that their method, normalizing sinking events per parent group, shows comparable sinking rates for central subhalos and level-1 satellites at high redshifts.

Page 10: The page discusses the radial distribution of subhalo sinking events within host halos, distinguishing between central-satellite and satellite-satellite sinking dynamics. Central-satellite sinking events show a centrally concentrated profile with peaks at $r \leq 0.1R_{\text{vir}}$ but can also occur at large radii due to spatial dispersion effects, while satellite-satellite sinking events preferentially occur in the outer halo regions across different redshifts.

Page 11: Figures 9 and 10 depict stacked radial probability functions and decomposed cross-level sinking types for subhalos in host halos with mass $[10^{12}, 10^{13}] M_{\odot} h^{-1}$ across various redshifts. The figures illustrate the normalized radial distances of sinking events relative to R_{vir} at merger, distinguishing between next-level and cross-level mergers, as well as different types of level-3 subhalo sinking events into level-1 subhalos.

Page 12: The radial distribution of sinking subhalos is consistent across hierarchical levels for direct parent-child mergers but flattens distinctly for level-3 to level-1 mergers, dominated by Orphaned Sinking events. This behavior is relatively stable over cosmic time, highlighting the role of hierarchical structure formation in subhalo coalescence locations. The sinking algorithm identifies physically resolved merger phases, with core scales (σ_x, σ_v) dependent on subhalo mass and numerical resolution, showing that higher resolution can affect both spatial and velocity dispersions systematically.

Page 13: The page presents Figure 11, showing spatial and velocity dispersions for central and satellite subhalos as functions of their bound mass, indicating how these properties relate to simulation softening lengths and a fitting formula from previous work. Additionally, Figure 12 illustrates the orbital trajectory of a level-1 subhalo in both xy- and yz-plane projections, along with its evolution in specific angular momentum, radial distance, and tangential velocity over time, highlighting significant orbital decay since infall.

Page 14: The page discusses the classification of subhalos into sunken and orbiting populations based on their minimum distance values in a 2D plane defined by δs_p and δs_c . It highlights that resolved subhalos exhibit three distinct sub-populations: a sunken component, and an orbiting component split into two groups representing different stages of orbital evolution influenced by dynamical friction. Unresolved subhalos show only two sub-populations, with their distribution peaking between 2 and 10, indicating significant dynamical friction influence before becoming unresolvable.

Page 15: The page presents Figure 14, which shows the distribution of minimum phase-space distances for sunken dark matter subhalos at $z=0$, partitioned into three regions based on merger identifiability criteria. Additionally, Figure 15 illustrates the temporal evolution of δs_p and δs_c for three level-1 subhalos identified by the sinking algorithm, highlighting differences in distance measures relative to their direct parent and showing characteristic core scales over time.

Page 16: The refined algorithm identifies approximately 32% more sinking events than the original one, primarily due to child-dispersion-driven sinking, which accounts for about 24.2% of total sunken subhalos. Child-dispersion-driven sinking is mainly driven by tidal heating, causing significant disruption, while parent-dispersion-driven and doubly identified sinkings show less evidence of heating and are more influenced by orbital decay.

Page 17: Sunken subhalos identified by the group-level rate are predominantly major mergers due to strong dynamical friction, with this trend consistent across hierarchical levels and redshifts. The merger ratio decreases in dynamically younger systems, corresponding to higher redshift or larger mass halos, as lower mass ratio mergers occur earlier. Additionally, satellite-satellite mergers preferentially happen in the halo outskirts and are significantly influenced by orbital decay and tidal heating mechanisms.

Page 18: Figure 16 illustrates the distribution of spatial dispersion ratios before and after sinking events for both child and parent subhalos within massive halos ($M_{\text{vir}} \geq 10^{12} M_{\odot} h^{-1}$), highlighting differences in dispersion-driven dynamics across various regions as defined in the study.

Page 19: Page 19 of the document lists references pertinent to the study of dark matter subhalos and their interactions within larger structures, including recent work on tidal evolution of satellite galaxies in cosmological simulations, which provides context for understanding the sinking and merging dynamics discussed in the paper.

Selected figures

